

CSE 427: Machine Learning

Practice Sheet (Final exam topics)

Topic 1: Backpropagation derivation

Q1. For a feed-forward network with inputs X_1, X_2 , one hidden layer of three sigmoid neurons (nodes 3, 4, 5) and two linear output neurons (nodes 6, 7) with loss $L = \frac{1}{2}\{(\text{Ob}_6 - Y_6)^2 + (\text{Ob}_7 - Y_7)^2\}$, derive the weight update rule for W_{25} (X_2 to hidden node 5). Show every chain-rule step.

Q2. A 2-3-2 network uses ReLU in the hidden layer and sigmoid in the output layer. With the same squared-error loss, derive W_{13}^{new} in terms of $\delta_6, \delta_7, W_{36}, W_{37}, \text{Ob}_6, \text{Ob}_7$, the ReLU derivative and X_1 . State the convention used for the ReLU derivative at zero.

Topic 2: Backpropagation numerical update

Q3. A 2-3-2 network has inputs $X_1 = 2, X_2 = 1$. All link weights are 0.5, hidden biases -0.2 , output biases -1.5 . Hidden uses ReLU, output is linear. Targets $Y_6 = 1, Y_7 = 0, \eta = 0.1$. Find the updated W_{14} and W_{37} .

Q4. A 2-4-3 network has $X_1 = 1, X_2 = 2$. All weights = 0.8, hidden biases = 1.0, output biases = 0.5. Hidden uses sigmoid, output is linear. Targets $Y_7 = 0, Y_8 = 1, Y_9 = 1, \eta = 0.1$. Compute updated W_{24} and W_{48} .

Topic 3: CNN output sizes and parameter counts

Q5. An input image of size $227 \times 227 \times 3$ is convolved with 64 filters of size $11 \times 11 \times 3$, stride 4, padding 2. The output is then convolved with 192 filters of size $5 \times 5 \times 64$, stride 1, padding 2. The result is convolved with 384 filters of size $3 \times 3 \times 192$, stride 1, padding 1. Find the output dimensions and parameter count after each phase (include bias terms).

Q6. An input of size $100 \times 100 \times 3$ is convolved with 16 filters of size $11 \times 11 \times 3$, stride 3, padding 2. The output is then convolved with 32 filters of size $5 \times 5 \times 16$, stride 1, padding 2. Compute the output dimensions and parameter count after each phase.

Topic 4: Convolution as matrix multiplication

Q7. For the first convolution in Q5, express the operation as a multiplication of two matrices. State (i) the shape of the unrolled-input matrix and what its rows and columns represent, (ii) the shape of the filter matrix and what its rows and columns represent, (iii) the shape and meaning of the resulting matrix.

Q8. An input of $32 \times 32 \times 3$ is convolved with 6 filters of $5 \times 5 \times 3$, stride 1, padding 0. Express this as a matrix multiplication, give the dimensions of both matrices and the output matrix, and explain how to reshape the result back into the standard feature-map form.

Topic 5: ROC curve interpretation

Q9. A learner is tested on 8 positive and 12 negative examples. Its ROC curve (axes in %) passes through (0, 0), (0, 50), (25, 50), (25, 87.5), (50, 100), (100, 100) where $x = \text{FPR}$ and $y = \text{TPR}$. Write True or False and correct each false statement: (i) At (25, 50) four positives are correctly classified and three negatives are misclassified. (ii) From (25, 50) to (25, 87.5) three more positives are correctly classified. (iii) At (50, 100) all positives are correctly classified and half of the negatives are misclassified. (iv) At (0, 50) precision equals 1.

Q10. An ROC curve has 5 positive and 5 negative samples and passes through (0.2, 0.4), (0.4, 0.8), (0.6, 1.0). True or False (correct if false): (i) At (0.2, 0.4) two positives are classified correctly. (ii) At (0.4, 0.8) four positives are correctly classified and two negatives are misclassified. (iii) Moving from (0.4, 0.8) to (0.6, 1.0) one more positive and one more negative cross the threshold. (iv) At (0.6, 1.0) recall equals 1.

Topic 6: RNN forward pass and SSR

Q11. For the simple RNN $h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$, $y_t = W_{hy}h_t + b_y$, inputs $\{x_1, x_2, x_3\} = \{2, 4, 6\}$, targets $\{3.2, 6.5, 8.9\}$. $h_0 = 0$, $W_{xh} = 0.4$, $W_{hh} = 1.2$, $b_h = -0.1$, $W_{hy} = 9.0$, $b_y = -1.5$. Predict y_1, y_2, y_3 and compute SSR.

Q12. Same RNN form. Inputs $\{1, 2, 3\}$, targets $\{4.0, 7.5, 10.2\}$. $h_0 = 0.5$, $W_{xh} = 0.6$, $W_{hh} = 0.9$, $b_h = 0.0$, $W_{hy} = 8.0$, $b_y = -3.0$. Predict outputs for all three time steps and find SSR.

Topic 7: RNN unrolling, vanishing gradient, modified RNN

Q13. Unroll the simple RNN for three time steps and derive an expression for $\partial \text{SSR} / \partial W_{hh}$. Using your expression, explain why the gradient vanishes when $|W_{hh}| < 1$ and the sequence is long.

Q14. A modified RNN uses $h_t = \tanh(W_{xh}x_t + W_{hh}(\alpha h_{t-1}) + b_h)$, with scalar $0 \leq \alpha \leq 1$. Explain why, even with α close to 1, this still underperforms an LSTM. Give at least three concrete limitations.

Topic 8: Positional encoding

Q15. The 2nd word of a sentence has embedding $[0.10, 0.40, 0.55, 0.20]$ ($d_{\text{model}} = 4$). Using $\text{PE}_{(\text{pos}, 2i)} = \sin(\text{pos} / 10000^{2i/d})$ and $\text{PE}_{(\text{pos}, 2i+1)} = \cos(\text{pos} / 10000^{2i/d})$, compute the encoded vector after adding positional encoding.

Q16. The 5th word of a sentence has embedding $[0.62, -0.18, 0.30, 0.05]$. Compute the positional encoding for position 5 with $d_{\text{model}} = 4$ and give the final positionally-encoded vector.

Topic 9: Self-attention (encoder)

Q17. Positionally encoded embeddings: "Cats" = $[1.20, 0.40]$, "sleep" = $[-0.60, 0.90]$. Encoder $Q = [[1.0, 0.5], [-2.0, 2.0]]$, $K = [[-1.4, 0.6], [-1.2, 1.0]]$, $V = [[1.6, -1.0], [-0.4, -0.3]]$. Compute the updated embeddings of "Cats" and "sleep" after self-attention. Use scaling by $\sqrt{d_k}$ and softmax.

Q18. Positionally encoded inputs: "He" = $[0.80, 0.20]$, "runs" = $[-0.30, 1.10]$. Encoder $Q = [[0.9, -0.4], [-1.8, 1.5]]$, $K = [[-1.0, 0.7], [-0.9, 1.2]]$, $V = [[1.2, -0.8], [-0.5, -0.1]]$. Find the self-attention output for both tokens.

Topic 10: Cross-attention (encoder-decoder)

Q19. After self-attention, [SOS] in the decoder has embedding $[1.80, -0.95]$. Encoder outputs: $e_1 = [0.50, -0.70]$, $e_2 = [-0.20, 0.60]$. Cross-attention $Q = [[1.2, -0.2], [0.4, -0.9]]$, $K = [[-1.0, 0.4], [-1.3, -0.7]]$, $V = [[1.1, 0.5], [-1.0, -0.4]]$. Compute the updated embedding of [SOS] after cross-attention.

Q20. Decoder token "chol" has post-self-attention embedding $[-1.20, 1.40]$. Encoder outputs: $e_{\text{Lets}} = [0.90, 0.20]$, $e_{\text{go}} = [-0.50, 0.65]$. $Q = [[0.3, -1.0], [1.4, -0.3]]$, $K = [[-0.4, -0.7], [-1.0, 0.4]]$, $V = [[-1.1, 0.4], [1.0, 0.7]]$. Compute the cross-attention output for "chol".

Topic 11: Word embedding concepts

Q21. Why is word embedding necessary? Compare "bag of words" and "skip gram": state how each is constructed, what each captures and one limitation of each.

Q22. Explain why one-hot encoding is unsuitable as input to a transformer. Then briefly describe how skip-gram (Word2Vec) learns its embeddings, including the role of the context window.